

COMPUTER NETWORKS

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CHAPTER 2

The Physical Layer

Comprehensive Textual Notes

A complete, detail-by-detail walkthrough for pre-reading the chapter

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Chapter 2 — The Physical Layer: Overview

The **physical layer** is the foundation (lowest layer) on which all networks are built. It defines the electrical, radio, and optical characteristics needed to move raw **bits** from one machine to another, and the properties of the media over which they travel. **Nature imposes two fundamental limits on every channel**, and together they bound the achievable data rate: the **Nyquist limit** (for noiseless channels) and the **Shannon limit** (for noisy channels).

The chapter is organized around: (1) the **transmission media** themselves, divided into **guided media** (twisted pair, coax, power lines, fiber) and **unguided/wireless media** (radio, microwave, infrared, light, satellite); (2) the **theory** that links waveforms to bits (Fourier analysis, Nyquist, Shannon) and the practical techniques of **digital modulation** and **multiplexing**; and (3) four large real-world systems that provide network access: the **telephone network (PSTN)**, **cellular networks (1G–5G)**, **cable networks (HFC/DOCSIS)**, and **communication satellites (GEO/MEO/LEO)**. It closes by comparing access networks and reviewing **policy/regulation** at the physical layer.

Key framing idea: **bandwidth** has two meanings that recur throughout. In the physical/analog sense it is a range of frequencies measured in **hertz (Hz)** (named for **Heinrich Hertz**); in the networking sense it is a data rate measured in **bits/sec**. The two are linked by the modulation and coding techniques of Section 2.4.

2.1 Guided Transmission Media

Guided media confine the signal to a physical path. The chapter surveys magnetic media (as a baseline), twisted pair, coaxial cable, power lines, and fiber optics, comparing each on bandwidth, distance, cost, and attenuation.

2.1.1 Persistent Storage ("never underestimate the bandwidth of a station wagon full of tapes")

Physically shipping storage media can beat any network for raw **throughput**, though with terrible **latency**. Worked example: an **Ultrium tape** holds about **30 TB**; a box of **1000 tapes = 800 TB = 6.4 petabits**. A courier (e.g., FedEx) can deliver the box cross-country in about **24 hours**, giving an effective rate of roughly **70 Gbps** — and if the destination is only an hour away, the rate is far higher. The cost works out to about **half a cent per GB**, cheaper than most networks.

- The catch is **delay**: shipping is high-bandwidth but high-latency, so it suits bulk transfer, not interactive use.
- Scale example: **Amazon Snowmobile** is a shipping container (a truck) that moves up to **100 PB** per trip — used to migrate huge data sets to the cloud physically.
- Lesson: for very large one-time transfers, "sneakernet" / physical shipment can be the highest-bandwidth option available.

2.1.2 Twisted Pair

A **twisted pair** is two insulated copper wires (typically about **1 mm** thick) twisted together in a helix. The twisting matters: it makes the two wires radiate and pick up interference in opposite phases, so unwanted signals tend to **cancel** (a pair twisted this way radiates far less than two parallel wires). Signaling is **differential** — the receiver reads the **difference** between the two wires, rejecting common-mode noise.

- **Cat 5e cable** — contains **4 twisted pairs** inside one sheath. **100-Mbps Ethernet** uses **2 of the 4 pairs**; **1-Gbps Ethernet** uses **all 4 pairs** in both directions simultaneously.

Transmission directionality terms:

- **Simplex** — communication in **one direction only** (e.g., broadcast).
- **Half-duplex** — both directions, but **only one at a time** (e.g., walkie-talkie).
- **Full-duplex** — both directions **simultaneously**.

Categories of twisted pair run from **Cat 3** up through **Cat 8**, with higher categories supporting more bandwidth (more twists per meter, tighter manufacturing). Through **Cat 6** the cable is typically **UTP (Unshielded Twisted Pair)**; **Cat 7** adds **shielding** around individual pairs and the whole cable. **Cat 8** supports very high rates over short runs (about **30 m**), aimed at **data centers**; its two classes are **Class I (compatible with Cat 6A)** and **Class II (compatible with Cat 7A)**. Twisted pair is the workhorse of the telephone **local loop** and of in-building LAN wiring because it is cheap and well understood.

2.1.3 Coaxial Cable (coax)

Coax has better shielding and higher bandwidth than twisted pair, so it spans longer distances at higher speeds. Construction: a stiff central copper conductor, surrounded by insulating material, a braided outer conductor (shield), and an outer plastic jacket. Two common types:

- **50-ohm coax** — used for **digital** transmission.
- **75-ohm coax** — used for **analog** transmission and **cable television (CATV)**.
- Modern coax has a bandwidth approaching **6 GHz**. A single analog TV program occupies roughly **3.5 MHz**.
- Coax was historically the backbone of long-distance phone and early Ethernet; today it survives most visibly in the **cable-TV / cable-Internet** last mile.

2.1.4 Power Lines

Electric **power lines** already reach every building, so reusing them for data is attractive — no new wiring. Data signals (high frequency, low power) are **superimposed** on the **50–60 Hz** power signal (power is low-frequency, data is high-frequency, so they coexist). Used both for **utility telemetry** (the power company reading meters) and for **in-home networking** (plug a data adapter into any outlet).

- Practical in-home rates reach up to about **500 Mbps** over short distances.
- Challenges: power wiring was never designed for data — it is **electrically noisy**, has **impedance that varies** as appliances switch on/off, and signals attenuate; it was not built as a transmission line.

2.1.5 Fiber Optics

Fiber uses pulses of light to carry bits and is the medium of choice for high-bandwidth, long-haul links. The chapter frames fiber progress against **Moore's law**: advances in transmission technology have kept pace with (even outrun) advances in computing. A fiber's theoretical bandwidth is enormous — **well over 50,000 Gbps (50 Tbps)** — but **practical rates are nearer 100 Gbps**, limited not by the glass but by our inability to convert between **electrical and optical (E/O) signals** faster.

An optical transmission system has three components: a **light source** (LED or semiconductor laser that turns an electrical pulse into a light pulse), the **transmission medium** (ultra-thin glass fiber), and a **detector** (a

photodiode that turns light back into electricity). Light is kept inside the fiber by **total internal reflection**: a ray striking the core/cladding boundary above a critical angle is reflected entirely back into the core and propagates with (almost) no loss.

Multimode vs. single-mode fiber:

- **Multimode fiber** — core wider than about **50 μm** (roughly a human hair); light bounces at many angles (many "modes"). Used for shorter runs, up to about **15 km**.
- **Single-mode fiber** — core only about **8–10 μm** , near the wavelength of light, so light travels in a straight line (one mode). It carries signals about **50 $\times$ farther** than multimode — e.g., **100 Gbps over 100 km** without amplification.

Attenuation is measured in **decibels per kilometer (dB/km)**; recall a factor-of-two power loss equals **3 dB**. Fiber loss is lowest in three **bands** centered near **0.85 μm , 1.30 μm , and 1.55 μm** ; each band is roughly **25,000–30,000 GHz** wide — colossal bandwidth. (The 0.85 μm band has higher loss but cheap optics; 1.30 and 1.55 μm have very low loss and are used for long haul.)

- **Chromatic dispersion** — different wavelengths travel at slightly different speeds, so a pulse **spreads out** over distance, limiting rate.
- **Solitons** — specially shaped pulses that counteract dispersion and can, in principle, travel without spreading.

Fiber cable construction: a glass **core** carries the light; it is surrounded by **cladding** (glass of lower refractive index to keep light in by total internal reflection), then a protective plastic **jacket**. Fibers are bundled into cables.

Joining fibers (three methods, by loss):

- **Connectors** — plug-in couplings; easy to reconfigure but lose about **10–20%** of the light (highest loss).
- **Mechanical splices** — two ends clamped/aligned; about **10%** loss.
- **Fusion splices** — ends melted together to form a solid joint; **lowest loss** (nearly seamless).

Light sources (Fig. 2-7 comparison): LEDs vs. semiconductor lasers. LEDs are cheaper, slower, longer-lived, less temperature-sensitive, and spread light over a wider spectrum; semiconductor lasers are more expensive, much faster, shorter-lived, more temperature-sensitive, and nearly single-wavelength. The **detector** is a **photodiode**, whose speed caps practical systems around **100 Gbps**.

Fiber vs. copper — why fiber wins for long haul:

- Repeater spacing: fiber needs repeaters only about every **50 km**, versus roughly every **5 km** for copper.
- Weight/size: **1000 twisted pairs running 1 km weigh about 8000 kg**; **two fibers** of higher capacity weigh about **100 kg** — far lighter and thinner.
- Fiber is immune to electrical interference, does not leak signal (hard to tap), and is not corroded by chemicals.
- Downsides: fiber is less familiar to install, requires skilled splicing, is harder to tap into (a plus for security, a minus for add/drop), and point-to-point interfaces are costlier.

2.2 Wireless Transmission

Wireless (unguided) media send signals through free space. A historical anchor: **ALOHA**, the pioneering wireless data network, was built by **Norman Abramson** in **Hawaii in the 1970s** — it introduced random-access ideas the chapter returns to later.

2.2.1 The Electromagnetic (EM) Spectrum

Moving electrons create electromagnetic waves that propagate through space (predicted by **James Clerk Maxwell** in **1865**, produced and observed by **Heinrich Hertz** in **1887**). A wave's **number of oscillations per second is its frequency f (in Hz)**; the distance between successive maxima is the **wavelength λ** .

- **Fundamental relation (Eq. 2-1)** — $\lambda f = c$, where $c \approx 3 \times 10^8$ m/s is the speed of light in vacuum (slower in copper/fiber, about $\frac{2}{3}$ of c).
- **Handy rule of thumb** — $\lambda f = 300$ when λ is in meters and f is in MHz (equivalently λ in metres $\times f$ in MHz = 300).

The usable spectrum runs from radio through microwave, infrared, and visible light into the ultraviolet, X-ray, and gamma regions (the latter are hard to generate/modulate and penetrate/harm living tissue, so they are not used for communication). The **ITU band names**, by increasing frequency, are **LF, MF, HF, VHF, UHF, SHF, EHF, and THF** (Low/Medium/High Frequency, then Very/Ultra/Super/Extremely/Tremendously High Frequency).

Why higher frequencies give more bandwidth: the amount of information a band can carry grows with its width in Hz. Worked example from the text: the fiber band at **1.30 μm** is about **0.17 μm wide**, which corresponds to roughly **30,000 GHz** of bandwidth — and at a modest **10 dB SNR** that is on the order of **300 Tbps** of capacity. This is why optical and millimeter-wave bands are so coveted.

Spread spectrum techniques deliberately spread a signal over a **wider** band than strictly necessary, trading bandwidth for robustness (resistance to interference/jamming and to narrowband noise), and allowing multiple users to share a band. Three variants follow.

2.2.2 Frequency Hopping Spread Spectrum (FHSS)

The transmitter **hops among many frequencies**, hundreds of times per second, in a pseudo-random sequence known to both ends. A narrowband jammer or interferer only corrupts the hops that happen to land on it, so the link is robust.

- Origin: co-invented by Hollywood actress **Hedy Lamarr** (with composer George Antheil); their design used **88 frequencies** (the number of keys on a piano) and received **U.S. patent 2,292,387** — intended to keep WWII torpedo-guidance signals from being jammed.
- Used by **Bluetooth** and by **older 802.11** wireless LANs; long a staple of **military** communication for its anti-jam, low-probability-of-intercept properties.

2.2.3 Direct Sequence Spread Spectrum (DSSS)

Each data bit is spread by multiplying it with a faster pseudo-random **code** (a chip sequence), spreading the energy across a wide band. Receivers that know the code can recover the signal even below the noise floor; those that don't see only noise.

- DSSS is the basis of **CDMA** (Code Division Multiple Access). It is used in **3G cellular**, in **GPS**, and in **802.11b** wireless LANs.

2.2.4 Ultra-Wideband (UWB) Communication

UWB sends a series of very short, low-energy pulses spread over an extremely wide band. By FCC definition a UWB signal occupies **at least 500 MHz** of bandwidth, or **at least 20% of its center frequency**. Because its energy is spread so thinly, it barely disturbs conventional narrowband users while delivering **hundreds of Mbps** over short ranges.

2.3 Using the Spectrum for Transmission

How a wireless signal behaves depends strongly on its frequency band. The chapter walks up the spectrum: radio, microwave, infrared, and light.

2.3.1 Radio Transmission

Radio waves are easy to generate, travel long distances, and penetrate buildings, so they are widely used. They are **omnidirectional** — they radiate in all directions, so the transmitter and receiver need not be precisely aligned.

- **Free-space path loss** — signal power falls off as $1/r^2$ with distance r — in free space the power drops about **6 dB each time the distance doubles**.
- **Low bands (VLF/LF/MF)** — propagate as **ground waves** that follow the earth's curvature, reaching roughly **1000 km** at low frequencies. **AM broadcast radio uses the MF band**. These bands pass through obstacles but carry little bandwidth.
- **Higher bands (HF/VHF)** — ground waves are absorbed, but these waves can **bounce off the ionosphere** (a layer of charged particles about 100–500 km up), enabling long skips. Amateur ("ham") and military short-wave radio exploit this.
- Interference example: radio is prone to interference from motors and other electrical gear — the text recounts GM **Cadillac** anti-lock brakes whose microprocessor emitted radio noise, and a case where **garage-door openers** shared bands with military radio.

2.3.2 Microwave Transmission

Above about **100 MHz**, waves travel in nearly **straight lines** and can be focused by a **parabolic antenna** (like a satellite dish) into a narrow beam, giving a high signal-to-noise ratio — but the transmitting and receiving antennas must be **accurately aligned**.

- **Line of sight / repeater spacing** — because microwaves travel straight and the earth curves, towers must periodically relay the signal. Repeater spacing grows with the **square root of the tower height** — e.g., **100-m towers** can be spaced about **80 km** apart.
- **Multipath fading** — some waves may be refracted off low atmospheric layers and arrive slightly delayed, out of phase with the direct beam; when they cancel, the signal fades. It is frequency-, weather-, and distance-dependent.
- **Water absorption** — microwaves are absorbed by water; around **4 GHz** absorption is a problem, and **rain** can disrupt links.

- Microwave was commercially pivotal: **MCI (Microwave Communications, Inc.)** built a long-distance business on microwave relay to challenge AT&T; **Sprint** grew from the **Southern Pacific Railroad**, which laid fiber along its rights-of-way. Microwave is cheap (no need to buy/dig land for cable — just put up towers) and still widely used for telephone and cellular backhaul.

2.3.3 Infrared Transmission

Infrared is used for **short-range** communication — the classic example is a **TV remote control**. It is cheap and easy to build, needs **no spectrum license**, and offers reasonable bandwidth.

- Key limitation: infrared **does not pass through solid objects** (walls), so it is confined to a room — a drawback for mobility but a **security plus** (signals can't be intercepted from outside the room, and rooms don't interfere with each other). The **IrDA** standard governed infrared data links.

2.3.4 Light Transmission (Free-Space Optics)

Unguided optical signaling has a long pedigree — the chapter cites **Paul Revere's lantern signals** ("one if by land, two if by sea") as early free-space optics. Modern **free-space optics** aims a **laser on one rooftop at a photodetector on another**, giving very high bandwidth at very low cost (no fiber to lay, no license needed).

- Downsides: laser beams are narrow and hard to aim; they cannot penetrate rain or thick fog; and **heat-driven air turbulence** bends the beam. The text recounts an anecdote where **rising convection currents** off a sun-heated building caused the laser beam to dance away from the detector — "convection currents can interfere with laser communication systems."
- **Visible Light Communication (VLC)** uses LED room lighting, flickered faster than the eye can see, to carry data — turning light fixtures into data transmitters.

2.4 From Waveforms to Bits

This section provides the theoretical core of the chapter: how continuous waveforms carry discrete bits, what fundamentally limits the rate, and the practical schemes (modulation and multiplexing) used in real systems.

2.4.1 The Theoretical Basis for Data Communication — Fourier Analysis

Any reasonably behaved periodic signal **$g(t)$** with period **T** (fundamental frequency **$f = 1/T$**) can be reconstructed as a sum of sines and cosines — a **Fourier series (Eq. 2-2)**:

$g(t) = c/2 + \sum a_n \sin(2\pi nft) + \sum b_n \cos(2\pi nft)$, where the sum runs over harmonics $n = 1, 2, 3, \dots$, the **a_n** and **b_n** are the sine/cosine amplitudes of the n th harmonic, and **$c/2$** is the DC (average) term.

Why this matters for communications: a transmission medium does not pass all harmonics equally. Real channels attenuate higher harmonics, and above a **cutoff frequency f_c** essentially nothing gets through. The more harmonics that survive, the more faithfully the received waveform resembles the one sent — so **limited bandwidth limits the data rate** and distorts sharp-edged digital signals.

- **Bandwidth** — the range of frequencies a medium passes (its width in Hz). Signals are **baseband** (frequencies from 0/near-DC up to a maximum, as sent on a wire) or **passband** (shifted up to sit around some carrier frequency, as in radio).

Worked example — the ASCII character "b" = 01100010. The text shows how this bit pattern, viewed as a periodic waveform, is built from its harmonics; as a channel passes more harmonics (1, 2, 4, 8...), the received

square wave becomes progressively sharper and more recognizable. A standard **voice-grade telephone line** passes roughly **3000 Hz** (about 300–3300 Hz).

Harmonics that fit in a 3000-Hz voice channel at various data rates (Fig. 2-13): at higher bit rates the fundamental frequency rises, so fewer harmonics fit under the 3000-Hz cutoff — which is why unaided voice lines top out at modest data rates.

Data rate (bps)	T (ms)	First harmonic (Hz)	# Harmonics sent (≤ 3000 Hz)
300	26.67	37.5	80
600	13.33	75	40
1200	6.67	150	20
2400	3.33	300	10
4800	1.67	600	5
9600	0.83	1200	2
19200	0.42	2400	1
38400	0.21	4800	0

Takeaway: **analog bandwidth (Hz)** and **digital bandwidth (bps)** are different but linked — to raise the bit rate on a fixed-Hz channel you must put more bits into each signal change (see modulation, 2.4.3), and there are hard physical limits (2.4.2).

2.4.2 The Maximum Data Rate of a Channel — Nyquist and Shannon

Two theorems bound every channel.

- **Nyquist (1924) — noiseless channel** — A noiseless channel of bandwidth **B** can carry at most **2B** signal changes (samples) per second. With **V** discrete signal levels, the maximum data rate is (**Eq. 2-3**) **max rate = $2B \log_2 V$ bits/sec**. So a noiseless 3-kHz channel cannot be sampled meaningfully faster than 6000 times/sec, but **more levels per symbol** buy more bits — in theory without limit, if there were no noise.
- **Shannon (1948) — noisy channel** — With noise present, the number of distinguishable levels is limited by the **signal-to-noise ratio S/N**. The maximum data rate is (**Eq. 2-4**) **max rate = $B \log_2(1 + S/N)$ bits/sec**. This is an absolute ceiling no coding scheme can beat.
- **SNR in decibels** — **$SNR(dB) = 10 \log_{10}(S/N)$** . (E.g., $S/N = 1000 \Rightarrow 30$ dB.)

Worked example: an **ADSL** channel of about **1 MHz** with a signal-to-noise ratio around **40 dB** ($S/N \approx 10,000$) gives a Shannon limit of roughly **13 Mbps** — in the right ballpark for real ADSL after overheads.

2.4.3 Digital Modulation

Digital modulation converts bits into signals suited to the channel. **Line codes** operate at **baseband** (on a wire); **passband** schemes shift bits onto a carrier for radio/optical channels. Good line codes balance **bandwidth efficiency**, **clock recovery** (enough transitions for the receiver to stay synchronized), and **DC balance** (equal time high and low, so no net charge builds up).

Baseband line codes:

- **NRZ (Non-Return-to-Zero)** — high voltage = 1, low = 0. Simple, but **long runs of identical bits have no transitions**, so the receiver can lose clock sync, and it is not DC-balanced.

- **NRZI (NRZ Inverted)** — a 1 causes a transition, a 0 causes none (or vice versa). Used by **USB**. Fixes long runs of 1s but not long runs of 0s.
- **Manchester encoding** — each bit has a mid-bit transition (combines clock + data), guaranteeing synchronization and DC balance, but needs **twice the bandwidth**. Used by **classic (10-Mbps) Ethernet**.
- **Bipolar / AMI (Alternate Mark Inversion)** — a 1 ("mark") is sent as alternating positive/negative pulses, a 0 ("space") as zero volts — keeps the signal DC-balanced. (Used in the T1 telephone system.)

Balancing codes (map data to slightly longer symbols to guarantee transitions / DC balance):

- **4B/5B** — maps every 4 data bits to a 5-bit symbol chosen so there are never too many consecutive 0s — a **25% overhead** (100 Mbps of data needs 125 Mbaud). Often combined with NRZI; a transition is guaranteed at least every few bit times.
- **8B/10B** — maps 8 data bits to 10-bit symbols (**80% efficiency**) and tracks **disparity** so that the running imbalance of 1s vs 0s stays within ± 2 — strong DC balance. (Used by Gigabit Ethernet, PCIe, etc.)
- **Scrambling** — XORs the data with a pseudo-random pattern to break up long runs without adding overhead bits (used where a little residual risk is acceptable).

Passband modulation (carrier-based) — vary one or more properties of a sinusoidal carrier:

- **ASK (Amplitude-Shift Keying)** — use two (or more) **amplitudes** to represent bits.
- **FSK (Frequency-Shift Keying)** — use two (or more) **frequencies**.
- **PSK (Phase-Shift Keying)** — shift the carrier **phase**. **BPSK** uses 2 phases (1 bit/symbol); **QPSK** uses 4 phases (**2 bits/symbol**).
- **QAM (Quadrature Amplitude Modulation)** — combine amplitude and phase to pack many bits per symbol: **QAM-16 = 4 bits/symbol**, **QAM-64 = 6 bits/symbol**, **QAM-256 = 8 bits/symbol**.
- **Constellation diagram** — plots each legal symbol as a point (amplitude = distance from origin, phase = angle). More points = more bits/symbol but points sit closer together, so they are **more vulnerable to noise**.
- **Gray code** — assigns bit patterns to adjacent constellation points so that neighbors differ by **only one bit** — a symbol error (slipping to an adjacent point) then causes just a single bit error.

2.4.4 Multiplexing

Multiplexing lets many signals share one medium. The main schemes:

- **FDM (Frequency Division Multiplexing)** — give each user a different **frequency sub-band**, separated by **guard bands**. Telephone voice channels are spaced **4000 Hz** apart (about 3100 Hz of voice plus guard band).
- **OFDM (Orthogonal FDM)** — split the band into many closely spaced **orthogonal subcarriers** whose spectra overlap without interfering, giving high efficiency and resistance to multipath. Implemented efficiently with the **FFT**. Used by **802.11 (Wi-Fi)**, **4G/LTE**, **DOCSIS 3.1 cable**, and **ADSL (as DMT)**.
- **TDM (Time Division Multiplexing)** — give each user the **whole band for a repeating time slot**. Requires synchronized clocks across the users. Underlies the digital telephone trunk system (T1/E1).
- **STDM (Statistical TDM)** — allocate slots **on demand** rather than in a fixed round-robin, so idle users waste no capacity — this is essentially **packet switching**.

CDM / CDMA (Code Division Multiple Access) — all users transmit in the **same band at the same time**, separated by orthogonal **codes**:

- Each user has a unique **chip sequence**; each data bit is transmitted as that whole sequence of **chips** (so the signal is spread — this is DSSS).
- Codes are **orthogonal**: for distinct sequences $\mathbf{S} \cdot \mathbf{T} = 0$, and a sequence with itself normalizes to $\mathbf{S} \cdot \mathbf{S} = 1$. A receiver recovers a user's bit by correlating the combined signal with that user's code; other users' contributions cancel to zero.
- Typical length $m = 64$ or **128 chips per bit** (e.g., **Walsh codes**). The text's worked example (Fig. 2-22) has four stations **A, B, C, D** transmitting simultaneously; the receiver dots the composite signal with a station's code to decode that station's bit.
- **WDM (Wavelength Division Multiplexing)** — FDM applied to **fiber** — different **wavelengths (colors)** of light carry separate channels on one fiber. Invented around **1990**. **DWDM (Dense WDM)** packs channels with spacing as tight as **12.5 GHz**, allowing hundreds of channels (e.g., **320 channels**); a 2019 example cited runs **160 channels at 16 Tbps** on one fiber.

2.5 The Public Switched Telephone Network (PSTN)

For wide-area connectivity, the telephone system is foundational. The "**last mile**" (the local connection to each home) is the perennial bottleneck; the **backbone** is high-capacity fiber.

2.5.1 Structure of the Telephone System

Invention: **Alexander Graham Bell patented the telephone in 1876**. The network evolved through three stages: (1) **fully interconnected** — a wire between every pair of phones, which is hopelessly unscalable (n^2 wires); (2) a **central switching office** — every phone connects to a switch that patches calls through; and (3) a **multi-level switching hierarchy** as the network grew.

Three major components of the modern phone system:

- **Local loops** — analog **twisted-pair** lines from each subscriber to the nearest switching office (the last mile).
- **Trunks** — high-bandwidth **digital fiber** links connecting switching offices to one another.
- **Switching offices** — where calls are routed from one line/trunk to another.

Hierarchy and terminology: each subscriber's local loop runs to a **local central office / end office** (the US has about **22,000** of them). Above end offices sit **toll (tandem) offices**, connected by **toll-connecting trunks**; toll offices connect to each other via **intertoll / interoffice trunks**. The classic Bell System used a **five-level hierarchy**.

- The **1984 AT&T breakup** transformed routing from a strict **hierarchical** scheme to a more flexible **non-hierarchical** one.

2.5.2 The Local Loop: Telephone Modems, ADSL, and Fiber

Modems. A **modem (modulator-demodulator)** converts the computer's digital bits to analog tones for the voice-grade loop and back. The usable voice band is about **3100 Hz**, so Nyquist caps the symbol rate near **6000 baud**; packing more bits per symbol (QAM) raised throughput. The **Shannon limit** on a plain voice line is about **35 kbps**.

Telephone modem standards (progression): **V.32 = 9600 bps**, **V.32 bis = 14,400 bps**, **V.34 = 28,800 bps**, **V.34 bis = 33,600 bps** (Fig. 2-27).

- **56k modems (V.90 / V.92)** — break the ~35 kbps barrier by exploiting the fact that one end (the ISP) connects **digitally** to the phone network, so only one analog conversion occurs. They run **56 kbps downstream** but only **33.6 kbps (V.90)** or **48 kbps (V.92)** upstream — asymmetric by physics.

PCM on the loop: a voice call is digitized at **8000 samples/sec** with **8 bits/sample**. In North America, one bit per sample is periodically "borrowed," leaving **56,000 user bits/sec** (hence the 56k target).

xDSL / ADSL — reusing the loop at its true bandwidth. The copper local loop can actually pass about **1 MHz**, far more than the 3100 Hz the phone company historically allowed; **DSL** exploits this (Fig. 2-28).

- **Channelization** — ADSL divides the ~1.1 MHz band into **256 channels of 4312.5 Hz each**, using **DMT (Discrete MultiTone)** — an OFDM scheme. **Channel 0 carries POTS** (ordinary phone service); **channels 1–5 are left unused** as a guard between voice and data.
- **Asymmetry** — more channels are assigned downstream than upstream (hence "A" = asymmetric), matching typical user demand. Each channel runs **QAM at about 4000 symbols/sec**, carrying up to **15 bits/symbol** on clean channels.

ADSL/VDSL standards and rates:

- **G.dmt (1999): ~8 Mbps down / 1 Mbps up.**
- **ADSL2: 12 Mbps / 1 Mbps.**
- **ADSL2+: 24 Mbps down**, using **2.2 MHz** of spectrum.
- **VDSL: 52 Mbps down / 3 Mbps up.**
- **VDSL2: 200 / 100 Mbps**, using up to **12 MHz**.
- **Vplus: 300 / 100 Mbps** over short loops (**under ~250 m**).

Equipment at the premises/office: a **NID (Network Interface Device)** marks the boundary of the phone company's property; a **splitter** separates voice (POTS) from data; at the central office a **DSLAM (Digital Subscriber Line Access Multiplexer)** aggregates many subscribers' data. Techniques like **bonding** (combining loops), **G.fast**, and **FTTdp** push rates higher as fiber moves closer to the home.

Fiber to the home — FTTX and PON. As fiber replaces copper on the loop, the family of architectures is called **FTTX**: **FTTH** (to the home), **FTTN** (to the node/neighborhood), **FTTC** (to the curb), and **FTTdp** (to the distribution point). The dominant access technology is the **PON (Passive Optical Network)** — passive optical splitters (no powered electronics) fan one fiber out to many homes.

- **GPON** — the ITU standard — about **2.4 Gbps downstream / 1.2–2.4 Gbps upstream**, shared.
- **EPON** — the IEEE (Ethernet) variant.
- **PONs reach up to about 20 km.** Because many homes share the fiber, downstream is broadcast (and encrypted); upstream uses **ranging** so each home knows its distance/timing and avoids collisions.

2.5.3 Trunks and Multiplexing

Trunks carry many calls digitally. The analog **FDM hierarchy** grouped **12 voice calls into a "group"** and **5 groups (60 calls) into a "supergroup."** Modern trunks are digital.

- **PCM digitization** — a **codec** samples voice **8000 times/sec** (one sample every **125 μs** — the Nyquist rate for a 4-kHz channel) at **8 bits/sample = 64 kbps** per voice channel. Two companding laws reduce quantization noise: **μ-law** (North America and Japan) and **A-law** (Europe), standardized in **ITU G.711**.

North American carrier hierarchy (T-carrier):

- **T1 = DS1 = 24 voice channels.** Frame = $24 \times 8 \text{ bits} + 1 \text{ framing bit} = 193 \text{ bits}$, every **125 μs** , giving **1.544 Mbps**. **Robbed-bit signaling** steals bits for control (about 8 kbps of signaling). Distinguishes **channel-associated** vs. **common-channel** signaling.
- **T2 = 6.312 Mbps, T3 = 44.736 Mbps, T4 = 274.176 Mbps** (each multiplexes lower levels).

European (ITU) carrier hierarchy: **E1 = 2.048 Mbps = 32 channels** (30 voice + 2 for framing/signaling); levels continue at **2.048 / 8.848 / 34.304 / 139.264 / 565.148 Mbps**.

SONET / SDH — the optical trunk standard. **SONET** (Synchronous Optical NETwork) came from **Bellcore in 1985**; the ITU's international version is **SDH** (Synchronous Digital Hierarchy, 1989, standards **G.707/708/709**). It was designed with four goals: carrier interworking, unifying US/European/Japanese systems, multiplexing digital channels, and OA&M (operations/administration/maintenance).

- **Synchronous:** all clocks are locked to a master reference accurate to about **1 part in 10^9** .
- **Frame:** an **810-byte frame sent every 125 μs** , drawn as **90 columns $\times$ 9 rows**. Overhead bytes split into **section, line, and path** overhead; user data rides in the **SPE (Synchronous Payload Envelope)**.
- **Base rate: STS-1 = 51.84 Mbps** (gross). Higher rates are integer multiples: **STS-1 $\rightarrow$ STS-768**. The optical signal for an STS-n is called **OC-n**. Concatenated channels (e.g., **OC-3c**) carry one payload rather than three independent ones.

SONET/SDH rate hierarchy (Fig. 2-35): Gross = $n \times 51.84$; SPE = $n \times 50.112$; User = $n \times 49.536$ Mbps.

SONET	SDH	Optical	Gross (Mbps)	SPE (Mbps)	User (Mbps)
STS-1	—	OC-1	51.84	50.112	49.536
STS-3	STM-1	OC-3	155.52	150.336	148.608
STS-9	STM-3	OC-9	466.56	451.008	445.824
STS-12	STM-4	OC-12	622.08	601.344	594.432
STS-18	STM-6	OC-18	933.12	902.016	891.648
STS-24	STM-8	OC-24	1244.16	1202.688	1188.864
STS-36	STM-12	OC-36	1866.24	1804.032	1783.296
STS-48	STM-16	OC-48	2488.32	2405.376	2377.728
STS-192	STM-64	OC-192	9953.28	9621.504	9510.912
STS-768	STM-256	OC-768	39813.12	38486.016	38043.648

2.5.4 Switching

The network distinguishes the **outside plant** (local loops, trunks — the cabling) from the **inside plant** (switches inside offices). Two switching paradigms:

- **Circuit switching** — a **dedicated physical path** is set up end-to-end for the whole call. Historical note: the **Strowger switch** was invented by an undertaker (**Almon Strowger**) who believed a human operator was diverting his calls to a rival — so he automated the operator away. Properties: a **call-setup delay** before any data flows; once set up, a **guaranteed constant rate with no congestion**; propagation adds about **5 ms per 1000 km**. Wasteful if the path sits idle (as data traffic often does).
- **Packet switching** — data is split into **packets**, each routed independently and **stored-and-forwarded** hop by hop. No setup delay and efficient statistical sharing, but packets experience variable **queueing**

delay and can arrive out of order; it is **fault tolerant** (reroutes around failures). This is how the Internet works.

Billing note: backbone links are often billed by the **95th-percentile** rule — usage is sampled in **5-minute intervals (8640 intervals over a 30-day month)**, the top 5% of samples are discarded, and the next-highest sets the bill. A comparison table (Fig. 2-38) contrasts circuit vs. packet switching on setup, path, congestion, and charging.

2.6 Cellular Networks

Mobile devices need wireless access the fixed phone system cannot provide. Cellular networks have evolved through five generations:

Gen	Era/character	Key technologies	Notes
1G	Analog voice	AMPS (FDM/FDD)	First-generation, analog only
2G	Digital voice	GSM, CDMA (IS-95), D-AMPS	GSM became dominant worldwide
3G	Digital voice + data	WCDMA/UMTS, CDMA2000	Based on broadband CDMA
4G	Packet-only (all-IP)	LTE, EPC (VoIP/VoLTE)	No circuit switching; voice over packets
5G	Very high rate / low latency	mmWave, massive MIMO, small cells	Up to ~20 Gbps; 1000× area capacity

2.6.1 Common Concepts: Cells, Frequency Reuse, Handoff

The core idea is to divide a region into **cells**, each served by a **base station**, and **reuse the same frequencies** in non-adjacent cells. Adjacent cells must use **different** frequency sets to avoid interference; a common pattern groups cells into **7-cell clusters** (Fig. 2-39), with the full set of frequencies reused once per cluster.

- **Capacity via smaller cells** — shrinking cells (**microcells**) reuses frequencies more often, multiplying capacity — the main lever for carrying more users in dense areas.
- **MSC / MTSO** — the **Mobile Switching Center (a.k.a. Mobile Telephone Switching Office)** coordinates all base stations in an area and links to the PSTN.
- **Handoff** — as a phone moves between cells, the call is transferred from one base station to another — a **handoff** takes about **300 ms**.
- **Channel types** — cells use four logical channel categories: **control**, **paging**, **access**, and **data/traffic** channels.
- Siting anecdote: cell antennas are often placed on tall existing structures; the text notes **church steeples** (Catholic Church properties) being rented as antenna sites.

2.6.2 First-Generation (1G) Technology: Analog Voice

Early mobile telephony began with a **1946** car-phone service in **St. Louis** using **push-to-talk**. **IMTS (Improved Mobile Telephone Service)** followed — a powerful **200-W** transmitter with **23 channels** across **150–450 MHz**, but with very limited capacity (one big cell per city).

- **AMPS (Advanced Mobile Phone System)** — introduced by **Bell Labs in 1983**, deployed cellularly; **retired in 2008**. (The UK counterpart was **TACS**; Japan's was **MCS-L1**.) Cells were about **10–20 km** across.

- AMPS used **FDM** with **832 full-duplex channels** and **FDD (Frequency Division Duplex)**: **824–849 MHz** mobile→base (uplink) and **869–894 MHz** base→mobile (downlink), each channel **30 kHz** wide, giving roughly **45 voice channels per cell** after control overhead and reuse.

2.6.3 Second-Generation (2G) Technology: Digital Voice

2G digitized voice, improving capacity, quality, and security (encryption). The main 2G systems:

- **D-AMPS** — digital AMPS (standards **IS-54 / IS-136**), backward-compatible with analog AMPS.
- **GSM** — the globally **dominant** 2G system (detailed below).
- **CDMA (IS-95)** — Qualcomm’s code-division system — a different, influential approach that shaped 3G.
- **PCS (Personal Communications Services)** deployed 2G in the **1900 MHz** band in the US.

2.6.4 GSM — The Global System for Mobile Communications

GSM (originally **Groupe Spéciale Mobile**) launched in **1991** and became the world’s most widely deployed mobile system. Its defining user feature is the **SIM (Subscriber Identity Module)** card, which holds the subscriber’s identity and separates it from the handset.

Architecture (Fig. 2-40): handset → **air interface** → base station → **BSC (Base Station Controller)** → **MSC** → databases **VLR (Visitor Location Register)** and **HLR (Home Location Register)** → PSTN.

Radio parameters: GSM operates at **900 / 1800 / 1900 MHz** using **FDD** (uplink/downlink separated, e.g., +55 MHz). Channels are **200 kHz wide**, each shared by **8 users via TDM**. The 900-MHz band provides **124 simplex channel pairs**.

- **Frame structure**: a **148-bit data frame** is sent in a **577 μs** time slot (including a **30 μs guard** time; structured as **3 + 57 + 26 + 57 + 3 bits** = tail/data/training/data/tail). Eight slots form a TDM frame of **4.615 ms**.
- **Rates**: gross radio rate **270,833 bps**; after overhead each user gets about **24.7 kbps**, of which roughly **13 kbps** is the compressed speech.
- **Multiframes**: a **26-frame multiframe (120 ms)** carries traffic; a **51-frame multiframe** carries control. Control channels include **broadcast**, **dedicated**, and **common** (the common ones being **paging**, **random-access**, and **access-grant**).
- **MAHO — Mobile-Assisted HandOff** — the handset measures neighboring base stations’ signal strength and helps the network decide when and where to hand off.

2.6.5 Third-Generation (3G) Technology: Digital Voice and Data

Timeline marker: the **iPhone launched in 2007 (on a 2.5G network)** and gained **3G in 2008**. An interim step, **EDGE**, boosted 2G data rates. The ITU’s 3G vision, **IMT-2000 (1992)**, set targets of **2 Mbps stationary**, **384 kbps walking**, **144 kbps in a moving car**.

Two incompatible 3G families, both based on broadband CDMA:

- **WCDMA / UMTS** — from **Ericsson / Europe**, using **5 MHz** channels.
- **CDMA2000** — from **Qualcomm**, using **1.25 MHz** channels.
- WCDMA runs at **3.84 Mchips/sec** with spreading codes from **4 to 256 chips**; it uses **pseudorandom sequences** with good **cross-correlation and auto-correlation** properties, and very fast **power control (~1500 times/sec)** to combat the near-far problem.

Advantages of CDMA cellular: it exploits **silence** (voice is active only ~40% of a conversation); it lets **all cells use the same frequencies** (no frequency planning); it works well with **sectorized (e.g., 3-sector) antennas**; and it supports **soft handoff** (a phone talks to two base stations at once during transition) versus the abrupt **hard handoff** of TDMA systems (Fig. 2-43).

2.6.6 Fourth-Generation (4G) Technology: Packet Switching

4G abandoned circuit switching entirely — it is **packet-only (all-IP)**, carrying even voice as packets. The dominant standard is **LTE (Long Term Evolution)**, standardized by **3GPP** and meeting the ITU's **IMT-Advanced** requirements; a competing early technology was **WiMAX (802.16)**.

- **EPC (Evolved Packet Core)** — the all-IP core network that carries data and **VoIP**. Voice over LTE is **VoLTE**.
- Peak rates about **100 Mbps up / 50 Mbps down** (mobile); spectral efficiency targets of **15 / 6.75 bps/Hz** (down/up).
- Core elements: **eNodeB** (the LTE base station), **S-GW (Serving Gateway)**, **P-GW (Packet Data Network Gateway)**, **MME (Mobility Management Entity)**, and **HSS (Home Subscriber Server)**.

2.6.7 Fifth-Generation (5G) Technology

5G targets much **higher data rates (up to ~20 Gbps)**, **far lower latency**, and **~1000× the area capacity** of 4G. It achieves this through several techniques:

- **Ultra-densification** — many very small cells — **picocells (<100 m)** and **femtocells (tens of meters)** — to reuse spectrum intensely.
- **Millimeter waves** — use of **20–300 GHz** bands (wavelengths **under ~10 mm**); these carry huge bandwidth but travel short distances and need **directional antenna arrays** for beamforming.
- **Massive MIMO** — base stations with large arrays of antennas serving many users at once — **MU-MIMO (multi-user)** and **FD-MIMO (full-dimension)** — for spatial multiplexing.
- **Software-defined edge** — **network slicing**, **SDN (Software-Defined Networking)**, and **NFV (Network Functions Virtualization)** let operators carve the physical network into tailored virtual networks.

2.7 Cable Networks

Cable provides a third major access technology, grown from community TV antennas into a broadband Internet medium.

2.7.1 A History of Cable Networks: Community Antenna Television (CATV)

Cable TV was conceived in the **late 1940s** to serve rural and mountainous areas with poor broadcast reception. The scheme: put a **big antenna on a hilltop** to capture the broadcast signal, feed it into an amplifier called the **headend**, and distribute it over **coaxial cable** to subscribers' houses — hence **CATV (Community Antenna Television)**.

- It began as a "mom-and-pop" business; as subscribers grew, cables were spliced on and amplifiers added. Transmission was **one-way** (headend → users). By **1970** thousands of independent systems existed.

- In **1974 Time Inc. launched Home Box Office (HBO)** with cable-only movie content; other cable-only channels followed. This drove **consolidation** (big companies buying systems) and the need to **interconnect** distant systems — mirroring what happened in telephony 80 years earlier.

2.7.2 Broadband Internet Access Over Cable: HFC Networks

Over time, inter-city cable runs were replaced by high-bandwidth **fiber**, while coax remained for the last segment to homes. A system with **fiber for long-haul and coax to the house** is called **HFC (Hybrid Fiber-Coax)** — today's predominant cable architecture (Fig. 2-45a). The optical-to-electrical converters where fiber meets coax are **fiber nodes**; because fiber has so much more bandwidth, **one fiber node feeds many coax cables**.

A key architectural difference from telephone: in HFC, a single coax is **shared** among many houses in a neighborhood (contention possible); in the phone system, each house has its **own private local loop** (Fig. 2-45b).

Making TV and Internet coexist via FDM. In North America, cable-TV channels occupy **54–550 MHz** (with **FM radio at 88–108 MHz**); each channel is **6 MHz wide** (in Europe channels are **6–8 MHz** for PAL/SECAM and the low end starts around **65 MHz**). Modern cables run well above 550 MHz — often to **750 MHz or more**. The solution (Fig. 2-46):

- **Upstream** — data uses the **5–42 MHz** band (slightly higher in Europe) — the low, previously unused part of the spectrum.
- **Downstream** — data uses high frequencies **above the TV band (54 MHz and up)**.
- This creates a natural **asymmetry** (more spectrum above TV than below), which matches users' download-heavy demand. The upgraded, intelligent headend with a high-bandwidth fiber uplink to an ISP is called a **CMTS (Cable Modem Termination System)** — "CMTS" and "headend" refer to the same component. All **one-way amplifiers had to be replaced with two-way amplifiers** to support upstream.

2.7.3 DOCSIS

Cable data service runs over **DOCSIS (Data Over Cable Service Interface Specification)**, the **CableLabs** standard. Each cable modem uses one upstream and one downstream channel, allocated by **FDM**; a **6-MHz downstream channel with QAM-64 $\approx$ 36 Mbps gross ($\sim$ 27 Mbps net after overhead)**, rising to **$\sim$ 39 Mbps net with QAM-256** (European values are about $\frac{1}{3}$ larger due to wider channels). The modem-to-home side is typically **Ethernet** (often bundled with a Wi-Fi access point).

Version (year)	Headline capability
1.0 / 1.1 (1997)	Working limit $\sim$ 38 Mbps downstream / $\sim$ 9 Mbps upstream
2.0 (2001)	Tripled upstream bandwidth
3.0 (2006)	IPv6 support; channel bonding $\rightarrow$ hundreds of Mbps per home
3.1 (2013)	OFDM, wider channels, higher efficiency $\rightarrow$ over 1 Gbps downstream
3.1 Full Duplex (2017)	Multigigabit symmetric (equal up/down) capacity
DOCSIS Low Latency (2018)	Features to reduce latency

2.7.4 Resource Sharing in DOCSIS Networks: Nodes and Minislots

Because a coax segment is shared, the bandwidth any subscriber gets **depends on neighbors' usage** — one user streaming 8K video reduces what others can get (unlike a dedicated ADSL loop). Cable ISPs fight congestion

by **node splits**: dividing long cables and connecting each to its own fiber node, pushing fiber deeper and reducing homes-per-node.

- Node sizes shrank from **500–2000 homes** (≈ 2004) to **300–500 homes by 2019**; **N+0 HFC ("Fiber Deep")** can reach as few as **~ 70 homes**, eliminating cascaded amplifiers by running fiber nearly to the last coax segment.

Modem startup and ranging: a powered-up modem scans downstream channels for a periodic system-parameters packet, announces itself upstream, and is assigned upstream/downstream channels (reassignable for load balancing). Because upstream has more noise (the plant wasn't designed for data, and noise from many homes funnels to the headend), modems use conservative modulation from **QPSK up to QAM-128**, some symbols given to **trellis-coded** error protection — so upstream/downstream asymmetry is even larger than Fig. 2-46 suggests.

Upstream sharing with TDM minislots. Upstream time is divided into **minislots**; each subscriber transmits in a different minislot. The headend periodically announces a new round of minislots, but the announcement reaches modems at different times due to propagation delay — so each modem must know its **distance to the headend** via **ranging** (send a special packet, time the response). A typical minislot payload is **8 bytes**.

- To request bandwidth, a modem asks for minislots; if granted, the headend acknowledges which minislots are reserved. When **multiple modems share a request minislot**, **collisions** occur; a collided modem waits a **random time and retries, doubling the backoff after each failure** — this is **slotted ALOHA with binary exponential backoff**. (Ethernet's carrier sensing can't be used, since modems can't hear each other on the cable.) **CDMA** can let several modems share a minislot at reduced per-user rate.
- **Downstream** has only one sender (the headend), so there is **no contention and no minislots**; it uses a **fixed 204-byte packet** (including a **Reed-Solomon error-correcting code**), leaving a **184-byte payload** — chosen for compatibility with **MPEG-2** digital TV, so TV and data channels share one format.

2.8 Communication Satellites

Early experiments bounced signals off metallized balloons (too weak) and even off the **moon** (the US Navy built a working ship-to-shore moon-bounce system). The breakthrough was the **active satellite**, which **amplifies** the signal before retransmitting it. A communication satellite is essentially a **microwave repeater in the sky**, containing several **transponders** that each listen to a slice of spectrum, amplify it, and rebroadcast at a different frequency (to avoid self-interference) — a mode called a "**bent pipe**." Newer satellites add on-board digital processing (regeneration), which avoids amplifying uplink noise.

Orbital mechanics and placement. By **Kepler's law**, orbital period grows with radius to the **$3/2$ power**: near the surface the period is about **90 minutes**; at **35,800 km** it is **24 hours**; at **384,000 km** (the moon) about **1 month**. Satellites must avoid the two **Van Allen belts** (bands of charged particles that would destroy them), leaving **three safe orbital regions: GEO, MEO, and LEO** (Fig. 2-48).

Type	Altitude	Round-trip latency	Sats for global coverage
GEO	$\sim 35,800$ km	~ 270 ms	3
MEO	$\sim 5,000$ – $15,000$ km (between belts)	35–85 ms	~ 10
LEO	low (hundreds of km)	1–7 ms (idealized; ~ 40 – 150 ms in real deployments)	~ 50

2.8.1 Geostationary Satellites (GEO)

In **1945** science-fiction writer **Arthur C. Clarke** calculated that a satellite at **35,800 km** in a circular equatorial orbit would appear **motionless** in the sky (no tracking needed) and described a complete communication system — but deemed it impractical because fragile **vacuum-tube** amplifiers couldn't be put in orbit. The **transistor** changed that; the first active comsat, **Telstar**, launched in **July 1962**. These are **GEO (Geostationary Earth Orbit)** satellites.

- **Spacing:** GEO satellites should sit no closer than about **2° apart** in the 360° equatorial plane to avoid interference, so at most **180** can coexist; each transponder reuses capacity via multiple **frequencies and polarizations**. Slot allocation is managed (politically) by the **ITU**.
- **Satellites:** modern GEO satellites weigh **over 5000 kg**, draw several kW from solar panels, and use on-board rockets for **station keeping** (correcting drift). When fuel runs out (**~10 years**), the satellite is retired; its orbit eventually decays.

Satellite frequency bands (Fig. 2-49):

Band	Downlink	Uplink	Bandwidth	Problems
L	1.5 GHz	1.6 GHz	15 MHz	Low bandwidth; crowded
S	1.9 GHz	2.2 GHz	70 MHz	Low bandwidth; crowded
C	4.0 GHz	6.0 GHz	500 MHz	Terrestrial interference
Ku	11 GHz	14 GHz	500 MHz	Rain
Ka	20 GHz	30 GHz	3500 MHz	Rain, equipment cost

- **C band** was first for commercial use (now crowded, shared with terrestrial microwave). **L and S** bands were added by agreement in **2000** (narrow, crowded). **Ku** can space satellites as close as **1°** and reach **>500 Mbps**, but suffers from **rain** absorption (mitigated by widely separated ground stations). **Ka** has huge bandwidth but costly equipment.
- **Transponders and beams:** a modern satellite has about **40 transponders**, each typically **36 MHz** wide; channels are shared via **TDM and FDM**. Early satellites used one wide beam (a **footprint** covering $\sim\frac{1}{3}$ of the earth); modern ones use multiple **spot beams** (elliptical, as small as a few hundred km) to reuse spectrum. A US satellite might have one wide beam for the contiguous 48 states plus spot beams for Alaska and Hawaii.
- **VSAT (Very Small Aperture Terminal)** — low-cost microstations with **≤1 m** antennas (vs. 10 m for a standard GEO antenna) and about **1 W** output — uplink to **~1 Mbps**, downlink up to several Mbps. Because VSATs are too weak to talk to each other directly, a ground **hub** with a big antenna relays traffic between them (Fig. 2-50), trading extra delay for cheap terminals. Great for rural/developing areas — a solar-powered VSAT can "finish wiring the world."

2.8.2 Medium-Earth Orbit Satellites (MEO)

MEO satellites orbit **between the Van Allen belts**, drifting slowly (about **6 hours** to circle the earth) so they must be **tracked**. They have smaller footprints and need less power than GEO. Today they are used mainly for **navigation, not telecom**: the **GPS (Global Positioning System)** constellation of roughly **30 satellites at ~20,200 km** is the prime example.

2.8.3 Low-Earth Orbit Satellites (LEO)

LEO satellites are **close to the earth**, so ground stations need little power and **round-trip delay is small ($\approx 40\text{--}150\text{ ms}$ in real deployments)**, and launches are cheaper — but the satellites move fast, so **many are needed** and they must be tracked/handed off. Two classic voice constellations:

- **Iridium** — Motorola filed with the FCC in **1990** to launch **77** LEO satellites (element 77 = iridium); the design was revised to **66** satellites (would have made it "Dysprosium," element 66). Service began **November 1998**, went **bankrupt August 1999** (mobile phones had exploded in the meantime), and assets worth $\sim \$5$ billion were bought for $\sim \$25$ million. Restarted **March 2001** and still runs. Satellites orbit at **670 km** in **circular polar orbits**, arranged as **six "necklaces" (one satellite every 32° of latitude)** (Fig. 2-51); each has up to **48 spot-beam cells** and **3840 channels**. Distinctively, **calls are relayed satellite-to-satellite in space** (each satellite talks to four neighbors) before coming down (Fig. 2-52a).
- **Globalstar** — uses **48** LEO satellites but a traditional **bent-pipe** design — calls go **up to a satellite, back down to a large ground station**, then across terrestrial networks (Fig. 2-52b). This keeps complexity on the ground (easier to manage) and lets handsets be low-powered.

Small satellites and new constellations. Launches continue at roughly **20 per year**, including satellites over **5000 kg** — but also tiny ones. **CubeSats** (standard **$10\times 10\times 10\text{ cm}$** units, $\leq 1\text{ kg}$, launchable for as little as **$\$40,000$** as secondary payloads) make space research accessible, communicating mostly on **UHF/VHF**. **OneWeb** (and Boeing) aim to build **LEO satellite Internet backbones** (initially several hundred satellites, **Ku band**, using "**progressive pitch**" to avoid interfering with GEO satellites).

General properties of satellite links: even at light speed, GEO round-trip latency is **$250\text{--}300\text{ ms}$** (typically **$270\text{ ms}$** ; **$540\text{ ms}$** for a VSAT-via-hub path). By comparison, terrestrial **microwave $\approx 3\text{ }\mu\text{s/km}$** and **coax/fiber $\approx 5\text{ }\mu\text{s/km}$** (slower because EM waves move faster in air than in solids). Satellites are inherently **broadcast media** (sending to thousands costs the same as to one — a privacy problem solved only by **encryption**), their cost is **independent of distance**, they have **excellent error rates**, and they can be **deployed almost instantly** (vital for disaster/military response).

2.9 Comparing Different Access Networks

2.9.1 Terrestrial Access Networks: Cable, Fiber, and ADSL

Cable, FTTH, and ADSL are more alike than different — comparable service and, as they compete, comparable prices. All now use **fiber in the backbone**; they differ only in the **last-mile** technology.

- **Dedicated vs. shared capacity** — **Fiber and ADSL give each subscriber a dedicated connection**, so adding users barely affects existing ones and bandwidth is consistent. **Cable subscribers share a node**, so heavy use by some causes **congestion** for others — cable ISPs respond by **over-provisioning** and by **node splits**. DOCSIS 3.0 requires bonding **at least four channels ($\approx 170\text{ Mbps down} / 120\text{ Mbps up}$, $\sim 10\%$ signaling overhead)**.
- **Ceilings** — cable's ceiling is the **coax capacity**; **fiber has far more usable spectrum**. ADSL speed **drops sharply beyond a few km** from the central office, so rural reach needs fiber build-out (e.g., FTTN). The FCC's **Measuring Broadband America (MBA)** reports (annual) find ISPs generally meet advertised rates.
- **Reliability** — historically the phone/DSL network was more reliable than cable, but the gap narrowed — most cable and DSL now achieve at least **"two nines" (99% uptime)**; the traditional phone network reaches **"five nines"** (minutes of downtime per year). Satellite and metro-wireless are less reliable.

- **Security** — point-to-point **ADSL is inherently more secure** than shared cable, where any user can read all packets on the cable — so cable providers **encrypt** all traffic in both directions.

2.9.2 Satellites Versus Terrestrial Networks

Once imagined as the future, satellites were overtaken by terrestrial fiber after telecom **competition opened up in 1984 (US)** and later in Europe, when carriers replaced long-haul with fiber and added ADSL. Still, satellites hold **niche advantages fiber cannot match**:

- **Rapid deployment** — critical for military and disaster response. After the **December 2004 Sumatra earthquake/tsunami**, satellites restored first-responder communications within **24 hours** (operators like **Intelsat**, with 50+ satellites, can rent capacity anywhere).
- **Poorly-served areas** — at sea, in deserts, or sparse terrain where cellular/terrestrial reach is inadequate; **Iridium** covers the entire earth. **Indonesia** launched its own satellite rather than cable its **13,677 islands**.
- **Broadcasting** — one satellite transmission reaches thousands of ground stations at once (TV distribution, stock/commodity feeds, direct-to-home digital TV/radio).

US satellite Internet providers include **Hughes (DISH/EchoStar)** and **Viasat** (mostly GEO/MEO, some moving to LEO). The FCC's 2016 MBA report found satellite providers were among the few with **declining** performance (capped around **10 Mbps**) due to growing subscribership and limited bandwidth. Interest is rising for **in-flight Internet** and **LEO backbones (OneWeb, Boeing)**, though throughput (**~50 Mbps**) remains below terrestrial. Bottom line: the mainstream future is **terrestrial fiber + cellular**, with satellites for specialized uses — though **economics** could shift this if launch costs drop dramatically.

2.10 Policy at the Physical Layer

Physical-layer technology is shaped heavily by regulation and policy, especially around spectrum.

2.10.1 Spectrum Allocation

The central challenge is allocating the EM spectrum **efficiently and fairly**, since co-channel transmitters in the same area interfere. National and international agreements divide frequencies among radio, TV, mobile, telephone, police, maritime, navigation, military, and government users. The ITU-R body **WRC (World Radiocommunication Conference)** coordinates globally so multi-country devices can be built, but countries aren't bound by it — the **FCC** (US allocator) has sometimes rejected ITU-R recommendations.

Three historical methods to assign allocated spectrum to specific companies:

- **Beauty contest** — each carrier argues why it best serves the public interest and officials choose — prone to **bribery, corruption, nepotism**.
- **Lottery** — a random draw among applicants — lets uninterested firms (a "hamburger restaurant") win and **resell at a windfall**.
- **Auction** — sell to the highest bidder. The UK's **2000 3G auction** expected **~\$4 billion** but raised **~\$40 billion** — enriching governments but saddling carriers with debt.

Unlicensed spectrum. The alternative is to **not allocate at all** but cap transmit power so devices have short range and don't interfere. Governments set aside **ISM (Industrial, Scientific, Medical) bands** for unlicensed use (garage openers, cordless phones, wireless mice, etc.). The FCC caps ISM power (e.g., **1 W**) and requires spread-spectrum techniques. US networking bands (Fig. 2-53): the **900 MHz** band (early 802.11, now crowded), the **2.4 GHz** band (802.11b/g, Bluetooth; interfered with by microwave ovens/radar), and **5 GHz** including the U-

NII (Unlicensed National Information Infrastructure) bands (used by 802.11ac — lots of bandwidth, now very popular and crowded).

- Unlicensed bands have driven enormous innovation (Wi-Fi, Bluetooth). Newer ideas: **LTE-U** (LTE in unlicensed spectrum, which must back off politely around incumbents), and use of **white spaces** (allocated-but-locally-unused bands). In **2009** the FCC allowed unlicensed use of white spaces around **700 MHz** (freed by the **2010** analog-to-digital TV transition), provided devices detect nearby licensed users (e.g., wireless microphones).
- The FCC opened **57–64 GHz** for unlicensed use (2001) — a huge block (more than all other ISM bands combined); at **60 GHz oxygen absorbs the waves**, giving short range ideal for in-room high-def streaming. Many bands are being repurposed/auctioned for **5G — 2.5, 2.9, 3.5, 6, 24, 28, 37, 39, 49 GHz, the 3.7–4.2 GHz C-band, even 95 GHz**; the first **5G auction** launched in **late 2018** (the 24 and 28 GHz bands each had ~3,000 licenses).

2.10.2 The Cellular Network

Small political and marketing choices had outsized effects. The US adopted a **single analog system (AMPS)** mandated nationwide by the FCC, so a phone bought in California worked in New York. Europe initially let each country build its own system — a **fiasco** — then learned, and for **digital standardized on one system (GSM)** that works Europe-wide. By then the US had decided government shouldn't standardize and left digital to the market, producing **two incompatible digital systems** (plus minor ones).

- **Why Europe overtook the US in mobile adoption:** (1) one system usable everywhere/any provider; (2) **phone numbers** — in the US mobiles mix with fixed numbers so callers can't tell, leading carriers to make the **mobile owner pay for incoming calls** (which discouraged adoption), whereas Europe gave mobiles a distinct area code and kept "**caller pays**"; (3) widespread **prepaid** phones in Europe (up to **75%** in some areas), ideal for kids and light users with no monthly bill.
- **MVNOs (Mobile Virtual Network Operators)** — carriers that **own no infrastructure** but rent capacity from others (a "full" MVNO runs its own core). As cells shrink and small-cell hardware commoditizes (reference designs from Qualcomm, Intel), the network edge may **disaggregate** — "whitebox" eNodeBs plus virtual EPC (e.g., the Open Networking Foundation's **M-CORD** project).

2.10.3 The Telephone Network

Before 1984 the **Bell System** provided most US local and long-distance service. The government sued it as an illegal monopoly and won: on **January 1, 1984, AT&T was broken up** into **AT&T Long Lines, 23 BOCs (Bell Operating Companies)** grouped into **seven regional BOCs (RBOCs)**, and other pieces — by court order (the **MFJ, Modification of Final Judgment**). Result: more long-distance competition, better service, lower long-distance rates, but **higher local rates** (cross-subsidies ended).

- **New architecture:** the US was split into **164 LATAs (Local Access and Transport Areas)** (roughly area-code-sized). Within a LATA, a **LEC (Local Exchange Carrier)** held a local monopoly; **inter-LATA** traffic went to an **IXC (InterEXchange Carrier)** such as AT&T Long Lines (later Verizon, Sprint). Each IXC builds a **POP (Point of Presence)** in a LATA, and the LEC must connect every IXC to every end office on **identical terms** (Fig. 2-54), so subscribers can choose their long-distance carrier.
- **Telecommunications Act of 1996:** let cable, local phone, long-distance, and mobile companies enter each other's businesses, enabling bundled services and competition. It mandated **local number**

portability (keep your number when switching LECs; extended to mobile and fixed↔mobile in **2003**), which greatly increased competition.

- **Recent policy:** activity has shifted to ISPs. Two notable items: the insecurity of **SS7 (Signaling System 7)** — the protocol by which cellular networks interconnect — which Congress asked the FCC to address; and the classification of **SMS text messages as an "information service"** (unlike voice's "communications service"), subjecting them to different billing and privacy rules.

2.11 Summary

The physical layer is the basis of all networks. **Nature imposes two fundamental limits** that set channel bandwidth: the **Nyquist limit** (noiseless channels) and the **Shannon limit** (noisy channels).

Transmission media are **guided** (twisted pair, coaxial cable, fiber optics) or **unguided** (terrestrial radio, microwave, infrared, lasers through air, and satellites).

Digital modulation sends bits as analog signals: **line codes** operate at baseband, and signals are moved to a **passband** by modulating a carrier's amplitude, frequency, and/or phase. Channels are shared among users by **time, frequency, and code division multiplexing**.

The **telephone system** has three parts — **local loops, trunks, and switches**. **ADSL** reaches up to **~40 Mbps** on the local loop by running many parallel subcarriers; **PONs** bring fiber to the home for still-higher rates. Trunks carry digital data, multiplexed with **WDM** (many wavelengths per fiber) and **TDM** (many users per link); both **circuit and packet switching** play a role.

Cable networks evolved from coax to **HFC**, now offering up to **~1 Gbps** (soon ~10 Gbps), but with capacity **shared** among subscribers on a node. For mobility, the fixed phone system is unsuitable: cellular evolved from **1G analog (AMPS) → 2G digital (GSM dominant) → 3G broadband CDMA → 4G packet-switched core (all voice over packets) → 5G (small cells, massive MIMO, much more spectrum)**.

Finally, the physical layer is shaped not only by technology but by **policy** — standards bodies and regulators — most dynamically around **wireless spectrum**, where agencies keep **re-appropriating and auctioning** bands to meet growing demand for data.

Appendix — Quick-Reference Cheat Sheet

Key formulas

- $\lambda f = c$ ($c \approx 3 \times 10^8$ m/s); rule of thumb $\lambda(\text{m}) \times f(\text{MHz}) = 300$.
- **Fourier series:** $g(t) = c/2 + \sum a_n \sin(2\pi nft) + \sum b_n \cos(2\pi nft)$.
- **Nyquist (noiseless):** max rate = $2B \log_2 V$ bits/sec; max $2B$ symbols/sec.
- **Shannon (noisy):** max rate = $B \log_2(1 + S/N)$ bits/sec; $\text{SNR(dB)} = 10 \log_{10}(S/N)$.
- **Attenuation:** 3 dB = factor-of-2 power loss. **Free-space loss:** ~6 dB per doubling of distance (power $\propto 1/r^2$).

Bits per symbol

- BPSK = 1 · QPSK = 2 · QAM-16 = 4 · QAM-64 = 6 · QAM-256 = 8 bits/symbol.

Numbers worth remembering

- **Voice channel:** ~3100 Hz voice; spaced 4000 Hz in FDM. **PCM:** 8000 samples/s $\times$ 8 bits = 64 kbps; 125 μ s/sample.
- **T1** = 1.544 Mbps (24 ch, 193-bit frame); **E1** = 2.048 Mbps (32 ch). **SONET STS-1 / OC-1** = 51.84 Mbps.
- **ADSL:** 256 $\times$ 4312.5 Hz DMT channels over ~1.1 MHz; up to ~40 Mbps (ADSL2+ 24 Mbps). **GPON:** ~2.4 Gbps shared.
- **GSM:** 200 kHz channels, 8 users/TDM, 270,833 bps gross, ~13 kbps speech. **Handoff** ~300 ms.
- **Fiber bands:** 0.85 / 1.30 / 1.55 μ m, each ~25,000–30,000 GHz wide. Repeaters ~50 km (fiber) vs ~5 km (copper).
- **Satellites:** GEO 35,800 km / ~270 ms; MEO (GPS ~20,200 km) 35–85 ms; LEO (Iridium 670 km, 66 sats) ~40–150 ms.
- **Cable:** TV 54–550 MHz (6-MHz channels); upstream data 5–42 MHz. DOCSIS 3.1 $\rightarrow$ >1 Gbps downstream.